

TOPIC 4. Phase Diagrams I

- **Basic concepts:**
 - ✓ **Component, phase, micro constituent.**
- **Unary or one component phase diagrams.**
- **Binary isomorphous phase diagrams.**
- **Gibbs Phase Rule.**
- **Tie line and lever rule.**
- **Non equilibrium solidification.**
- **Binary eutectic systems.**

BASIC CONCEPTS

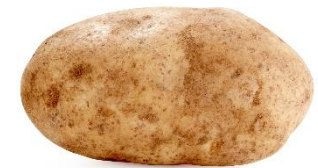
Component

Chemical substance or compound with a fixed composition that can form part of a phase or alloy. Examples: Aluminium (Al), Alumina (Al_2O_3).

Component	g/100 g (wet basis)
Water	77.23 ± 0.46
Protein	2.73 ± 0.36
Fat	0.30 ± 0.05
Ash	0.97 ± 0.18
Carbohydrates	18.77 ± 0.30
Reducing sugars	1.10 ± 0.01

Phase

Chemically and structurally homogeneous area that forms part of the microstructure. Could be formed by one or various components. Examples: Ferrite (solid solution Fe-C with BCC structure) austenite (solid solution Fe-C with FCC structure), calcium carbonate (CaCO_3)



Potato

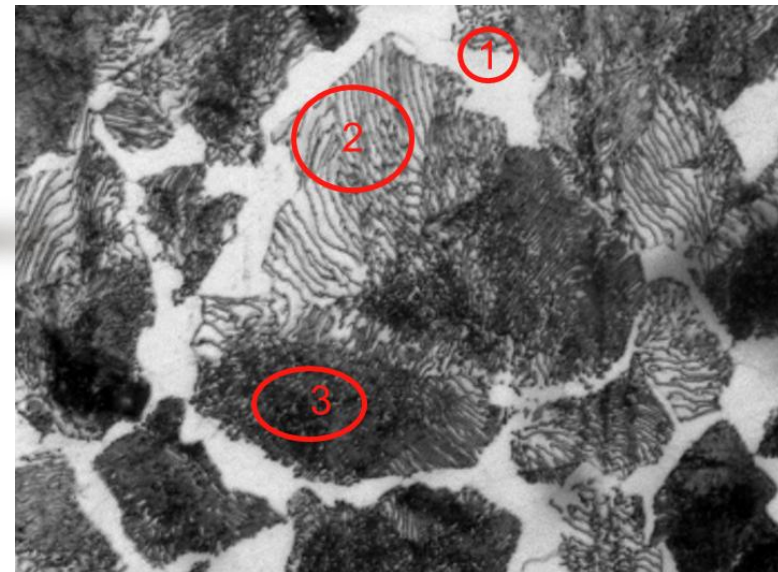
Microconstituent

Each one of different structures that can be observed on a polished surface of the material.

Can be formed by one or various phases



Fries



1: Ferrite

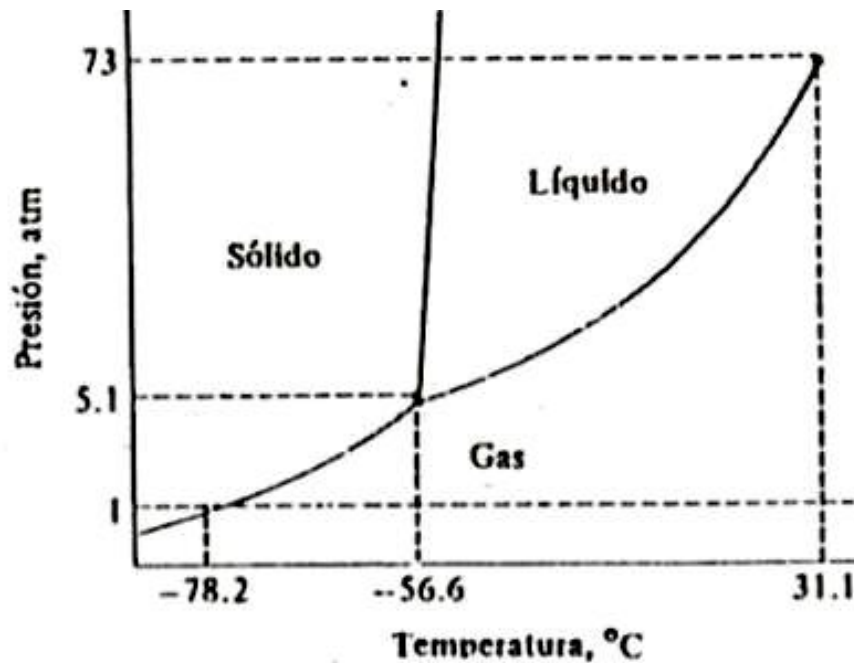
2 and 3: pearlite (ferrite + cementite)

DEFINITION OF EQUILIBRIUM PHASE DIAGRAM

Equilibrium diagram (aka Stability map) :

Map of the phases in thermodynamic equilibrium that are present in a system of materials at different pressures, temperatures and compositions.

Phase diagram of CO_2



- There is no liquid CO_2 at T_{room} and ambient pressure.
- Gradient of $P/T > 0$, as it is in almost all substances.
- Triple-point: gas, liquid and solid are at equilibrium.
- At T below the triple point CO_2 can sublime: gas $\rightarrow$ solid (dry ice)
- Above $T_c = 32^{\circ}\text{C}$ and $P_c = 73\text{ atm}$ $\rightarrow$ supercritical CO_2 fluid (used to extract caffeine from coffee)



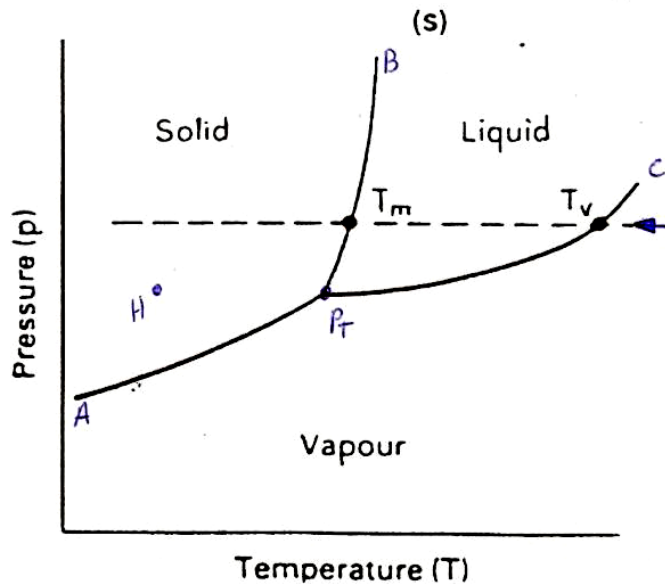
Dry ice or card ice: solid CO_2

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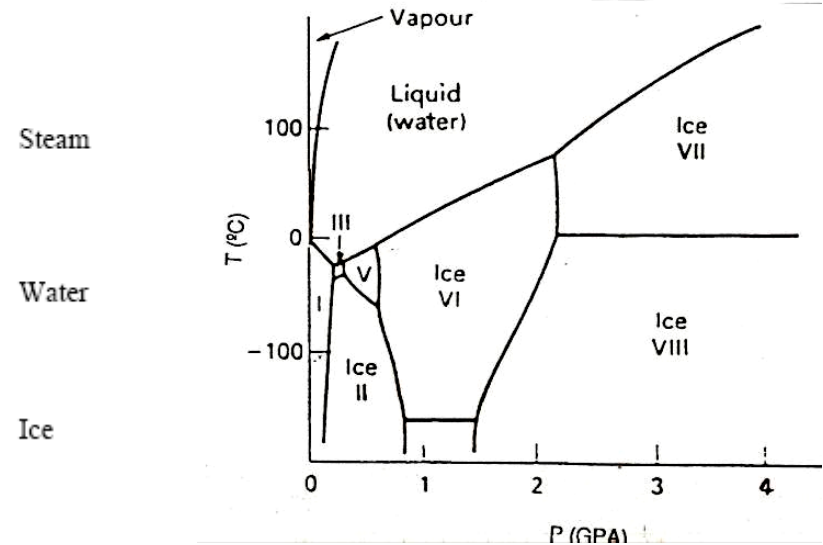
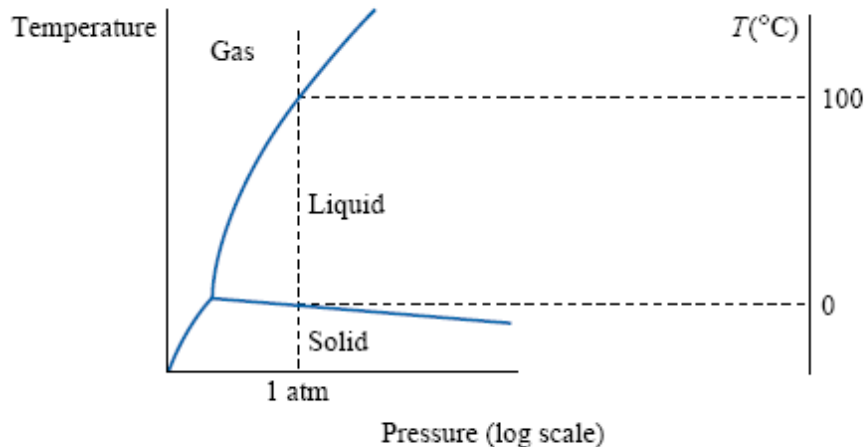
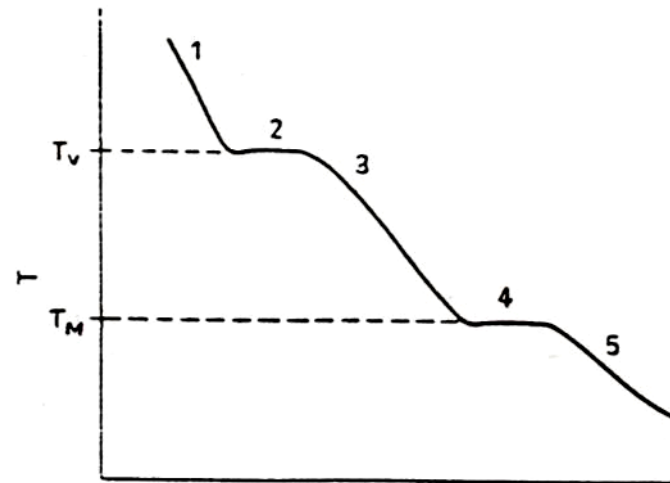
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- Binary eutectic systems.

UNARY OR ONE COMPONENT PHASE DIAGRAM : P-T

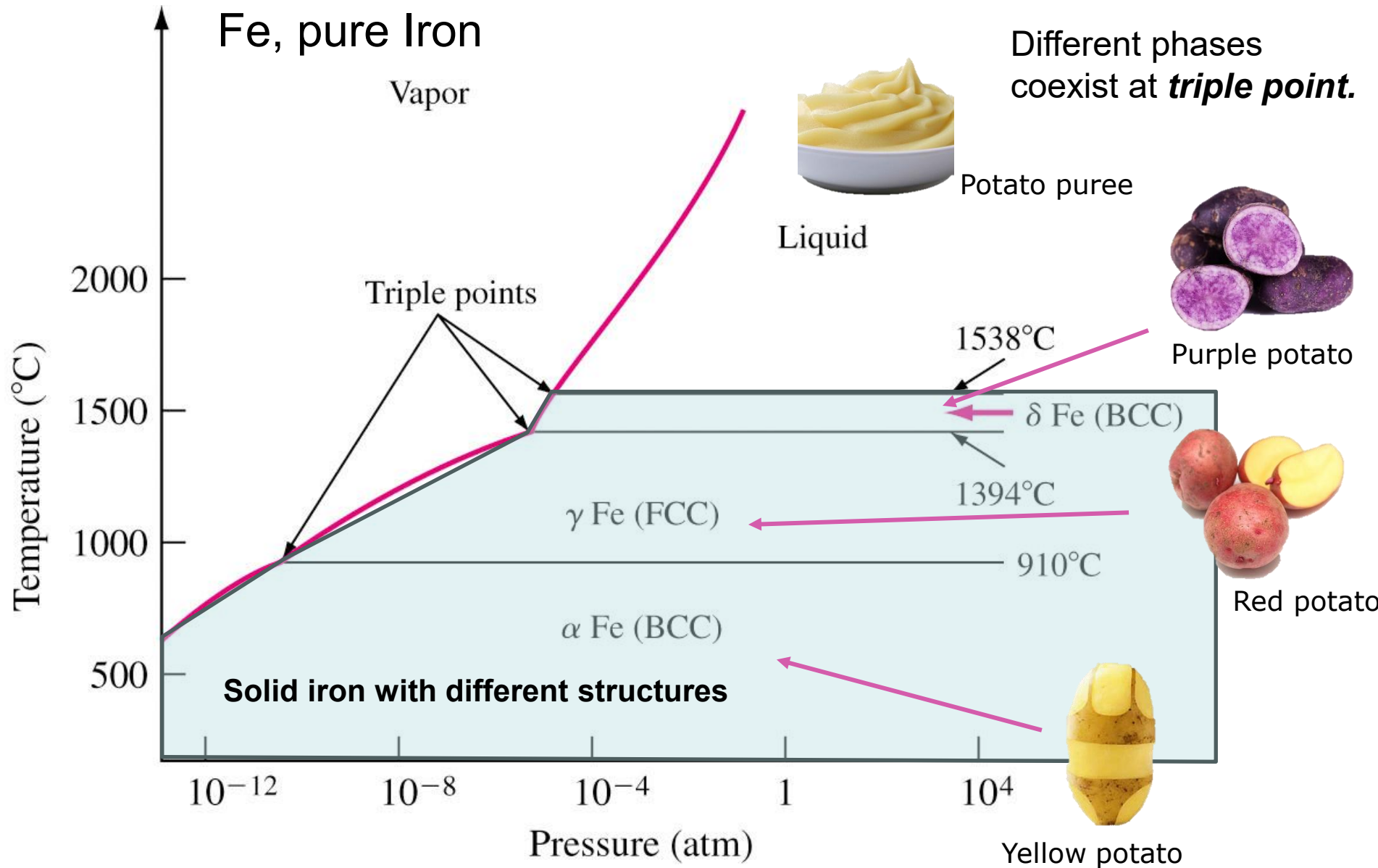
Phase diagram of water



Cooling curve for water



UNARY OR ONE COMPONENT PHASE DIAGRAM : P-T



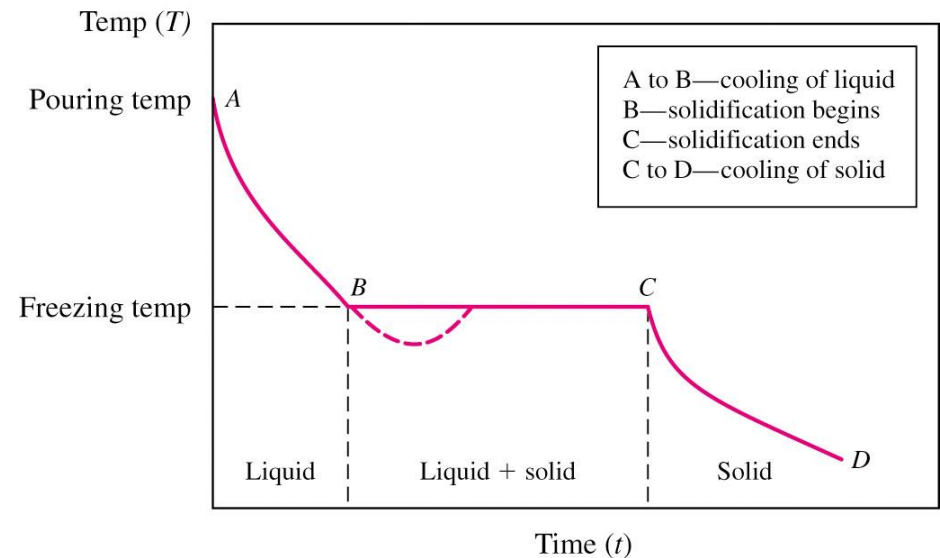
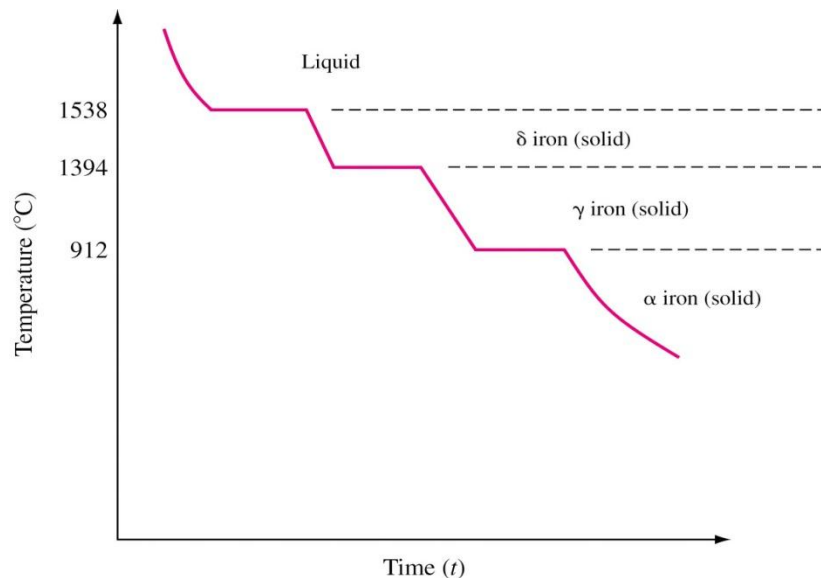
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BINARY ISOMORPHOUS PHASE DIAGRAMS

COOLING CURVES

- Used to determine **phase transition** temperature.
- Temperature and time data of cooling molten metal is recorded and plotted.

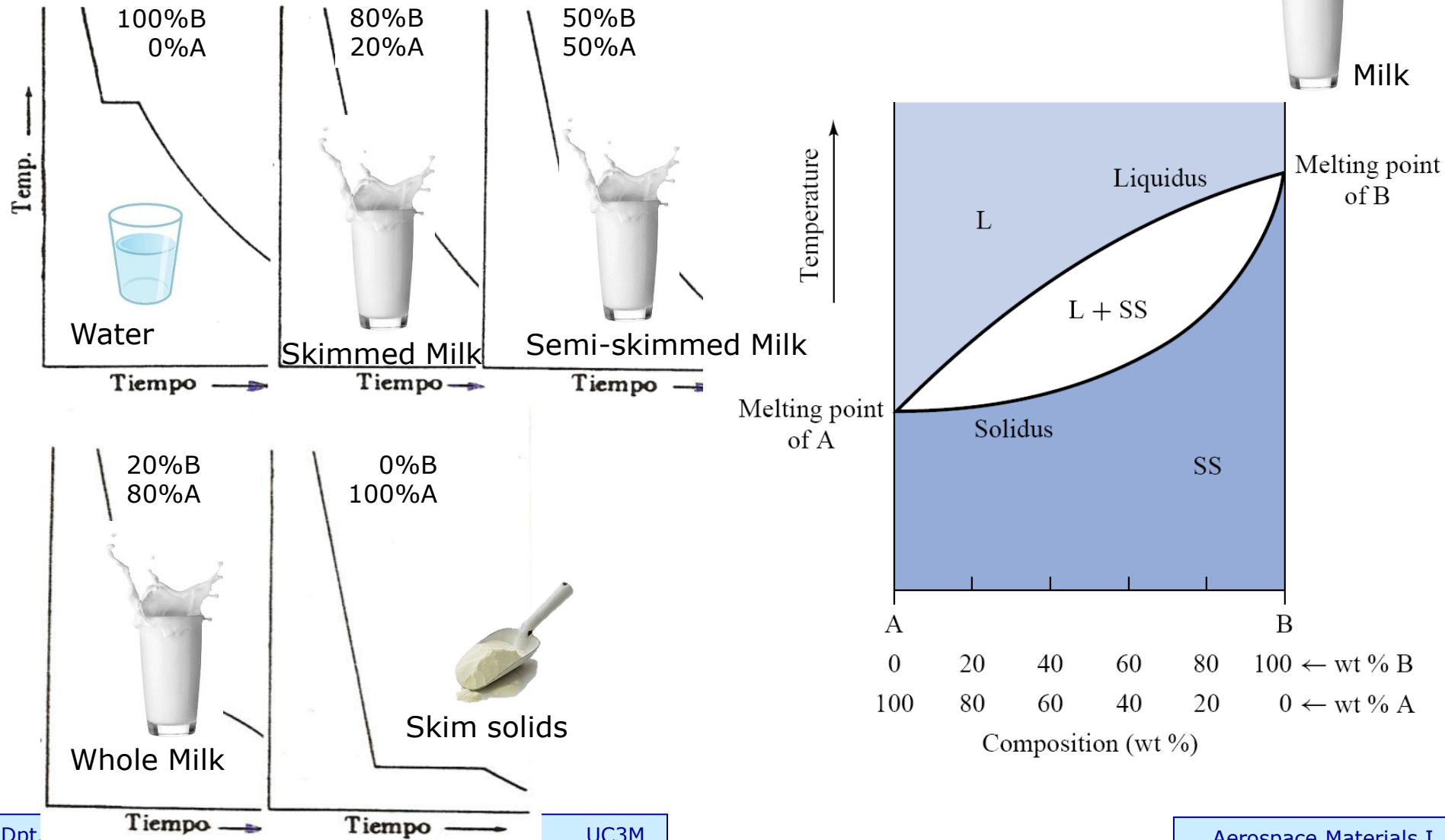


- Thermal arrest: heat lost = heat supplied by solidifying metal
- Alloys solidify over a range of temperature (no thermal arrest)

BINARY ISOMORPHOUS PHASE DIAGRAMS

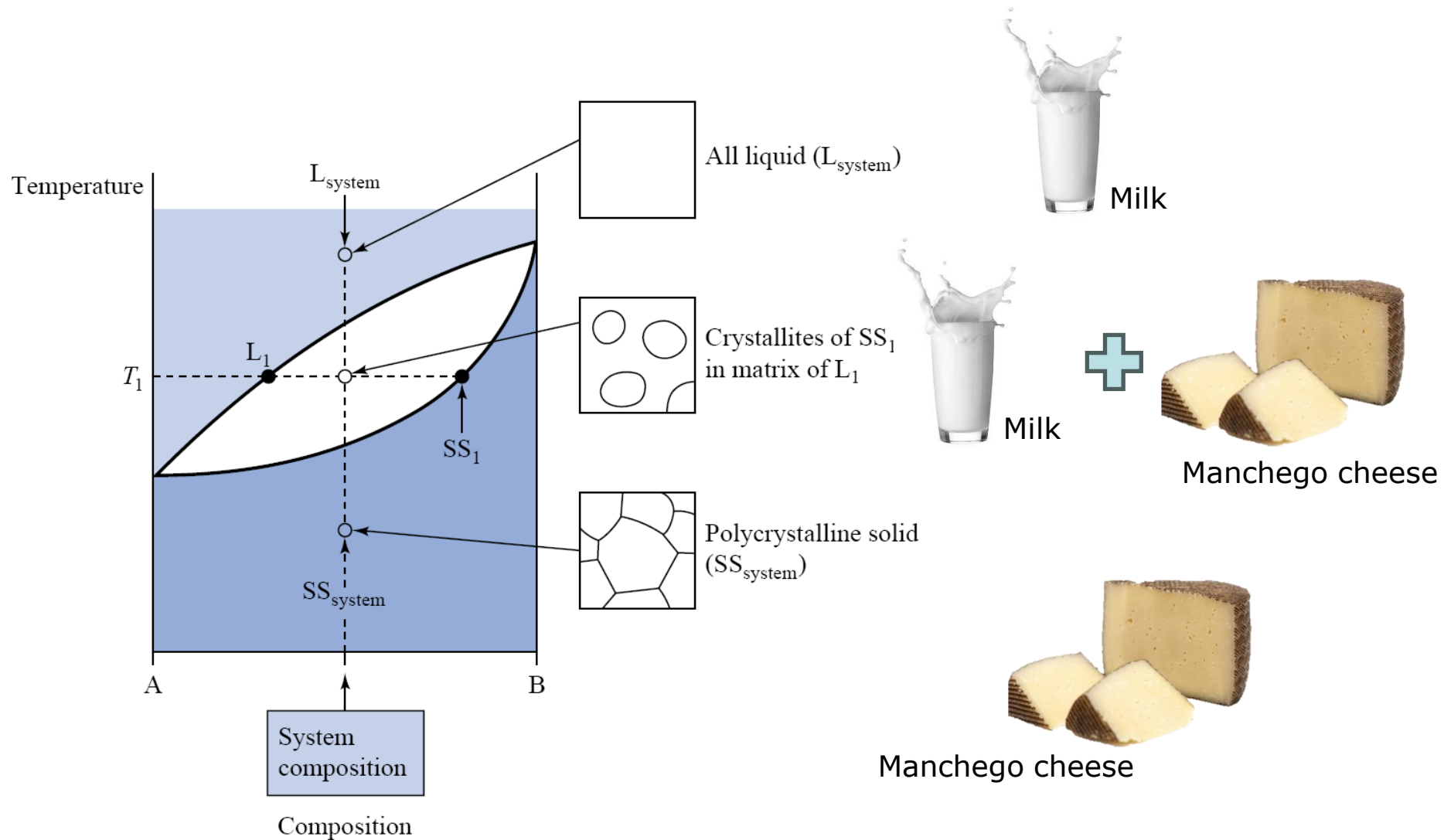
SYSTEMS OF 2 COMPONENTS WITH COMPLETE SOLID SOLUBILITY

Design of a phase diagram from the cooling curves



Milk

BINARY ISOMORPHOUS PHASE DIAGRAMS



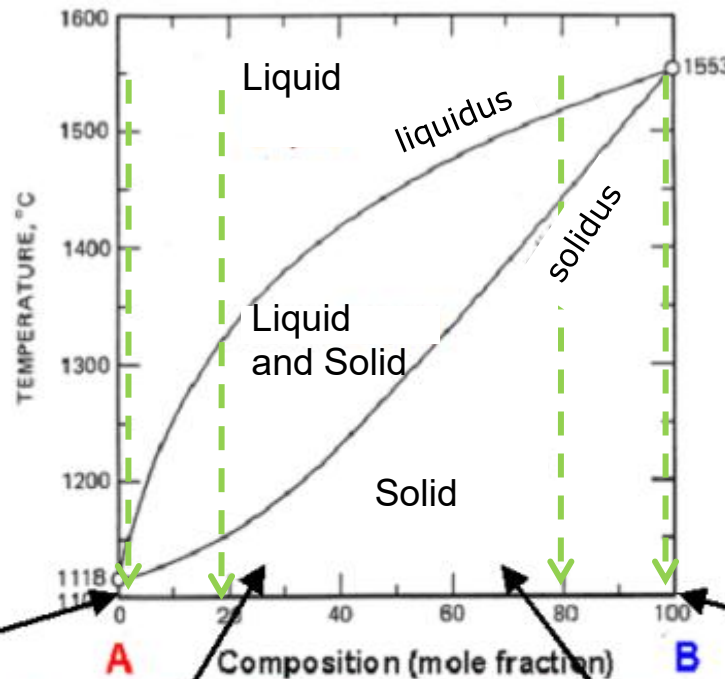
BINARY ISOMORPHOUS PHASE DIAGRAMS

Complete Solid Solubility

Polycrystalline single phase material.

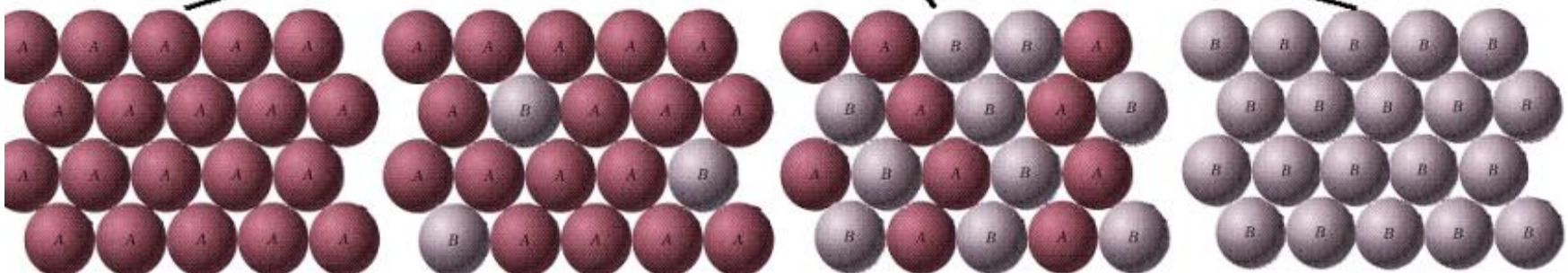
Its microstructure (*FCC, BCC, or any other*) is not different from that of the pure metal.

When solute is added, T_m is modified



Complete solubility in solid state.

The alloy **does not solidify at a constant temperature** but at a range of temperatures



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GIBBS PHASE RULE

$$F = C - P + N$$

F = Degrees of freedom (No. of independent variables available to the system)

C = No. of components

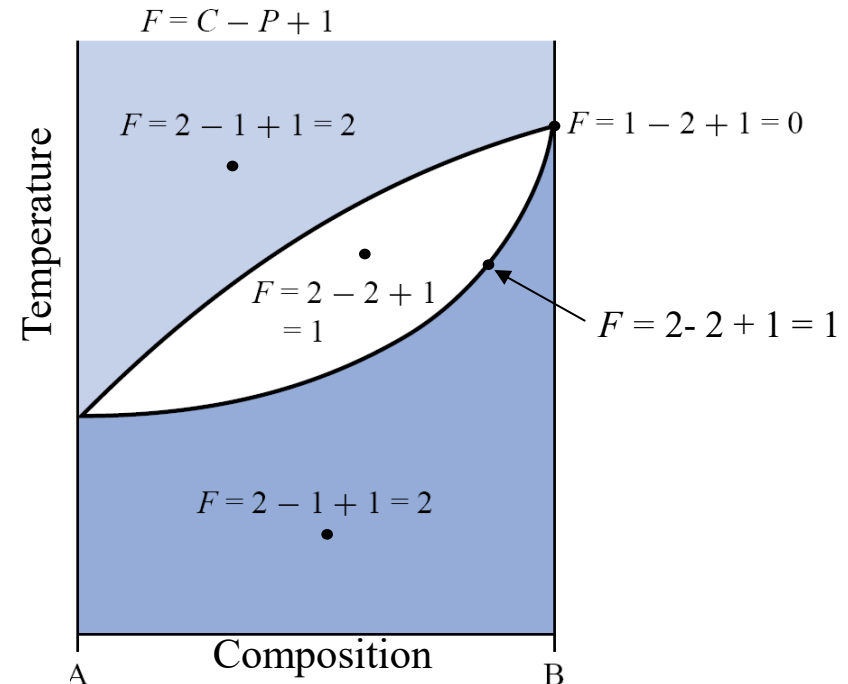
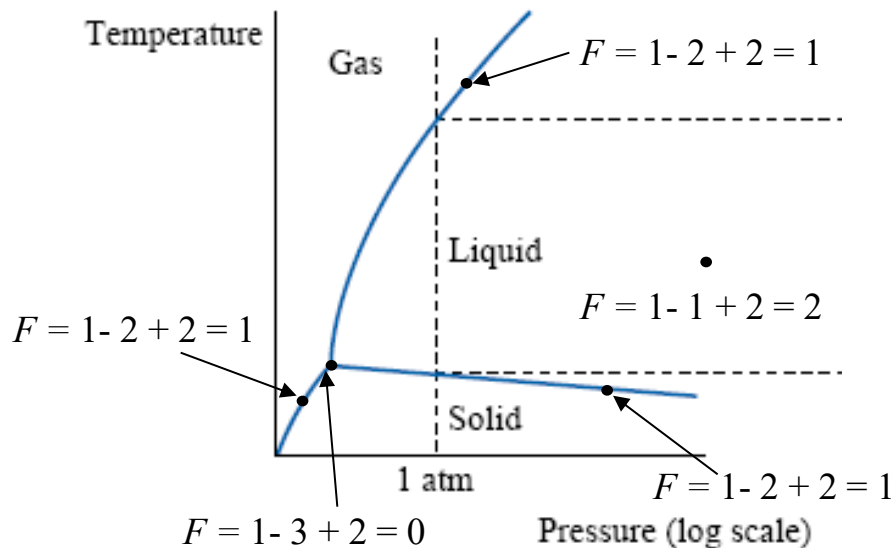
P = No. of phases

N = No. of noncompositional variables (e.g. pressure and temperature)

For routine processing of materials, we can consider the pressure to be fixed at 1atm:

$$\Rightarrow F = C - P + 1$$

$$F = C - P + 2$$

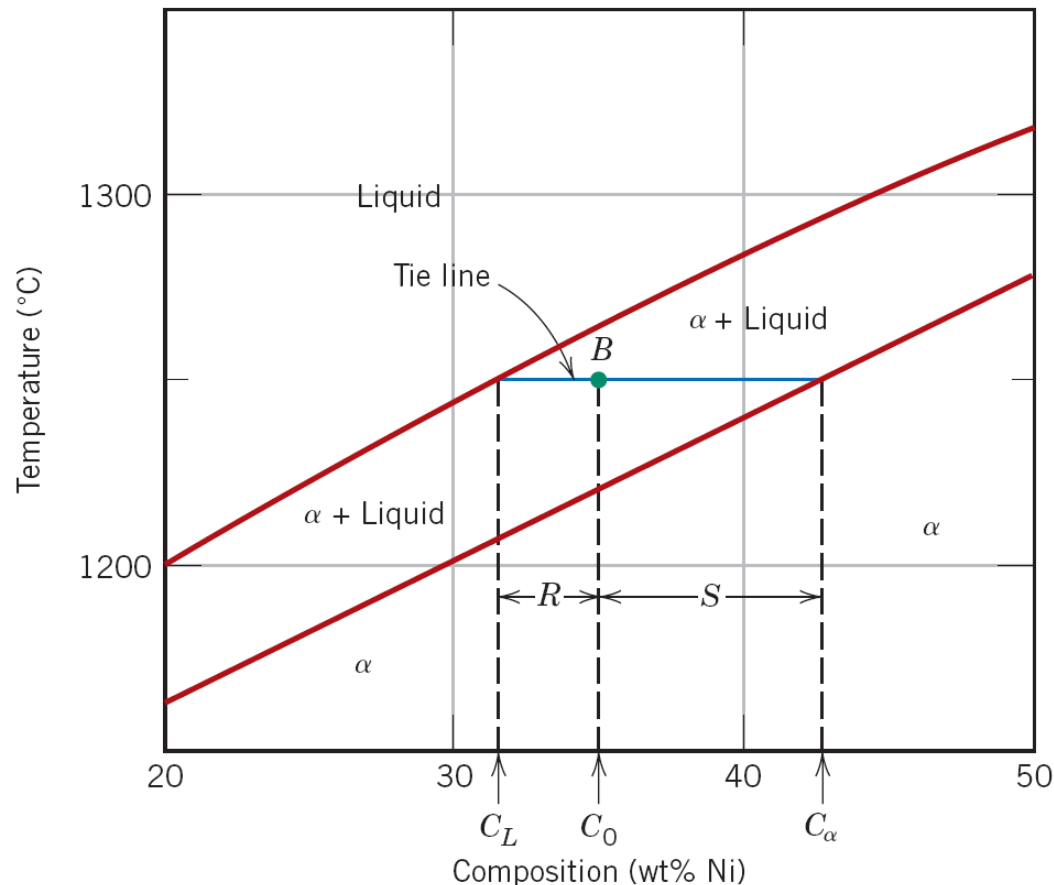


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TIE LINE AND LEVER RULE

- Determination of Phase composition: Tie line
- Determination of Phase Amounts: Lever rule



Phase composition:

- Intersections of the tie line with phase boundaries
- Composition of the respective phases is given by the composition axis

TIE LINE AND LEVER RULE

LEVER RULE

Determination of Phase Amounts

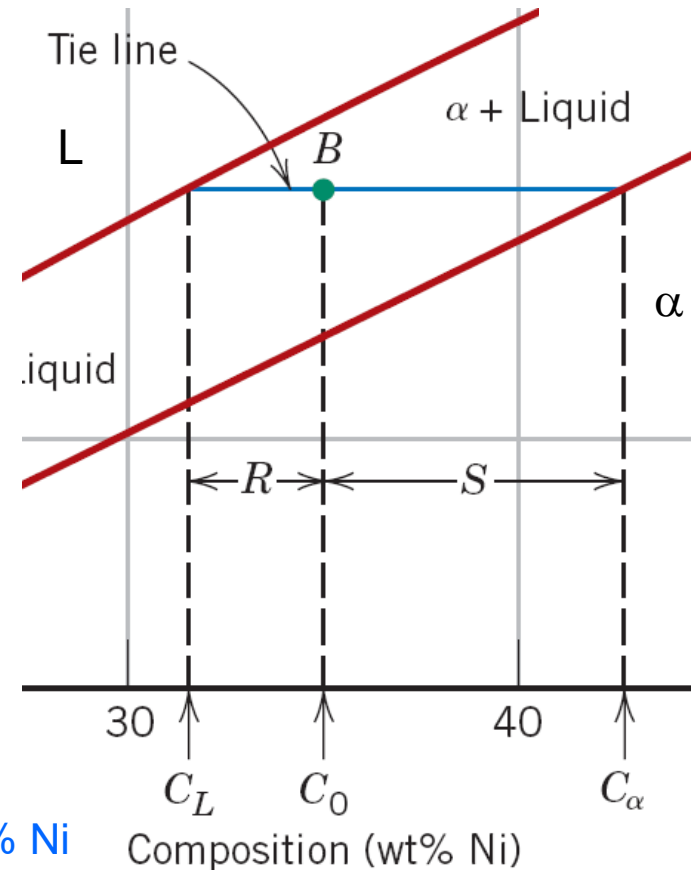
- The fraction of each phase is computed by taking the length of the tie line from the overall alloy composition to the phase boundary of the other phase, and dividing by the total line length

$$\boxed{W_L} = \frac{S}{R+S} = \frac{C_\alpha - C_0}{C_\alpha - C_L}$$

$$W_\alpha = \frac{R}{R+S} = \frac{C_0 - C_L}{C_\alpha - C_L}$$

W_L = fraction of liquid phase
 W_α = fraction of α phase

$C_L = 32 \text{ wt\% Ni}$
 $C_0 = 35 \text{ wt\% Ni}$
 $C_\alpha = 43 \text{ wt\% Ni}$



TIE LINE AND LEVER RULE

LEVER RULE

Determination of Phase Amounts

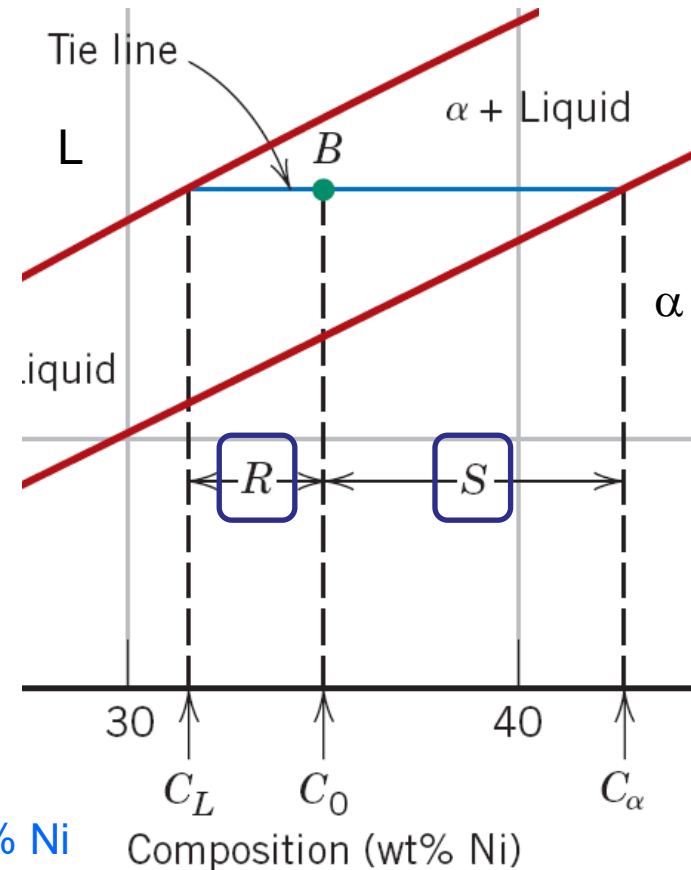
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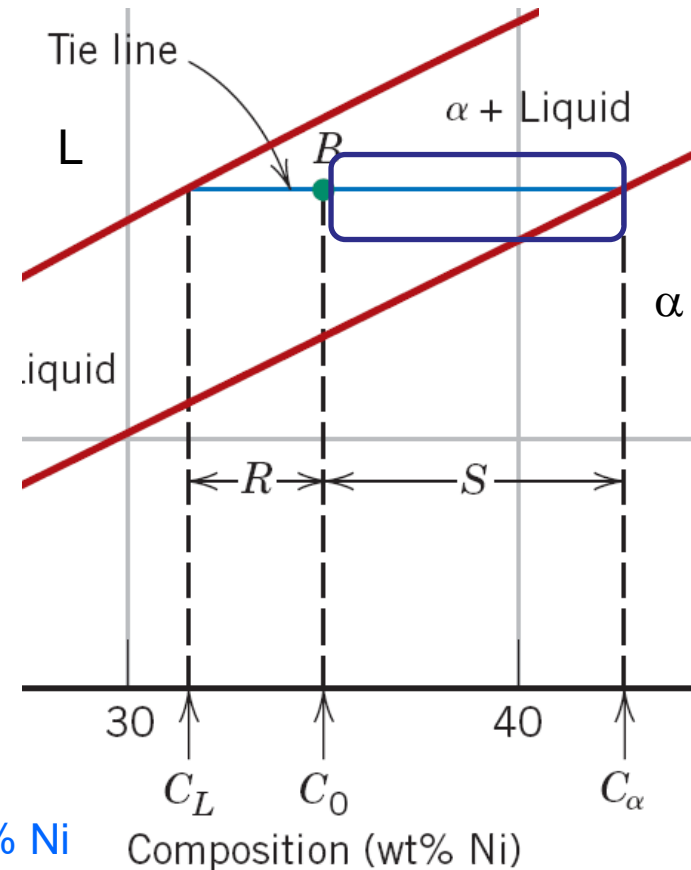
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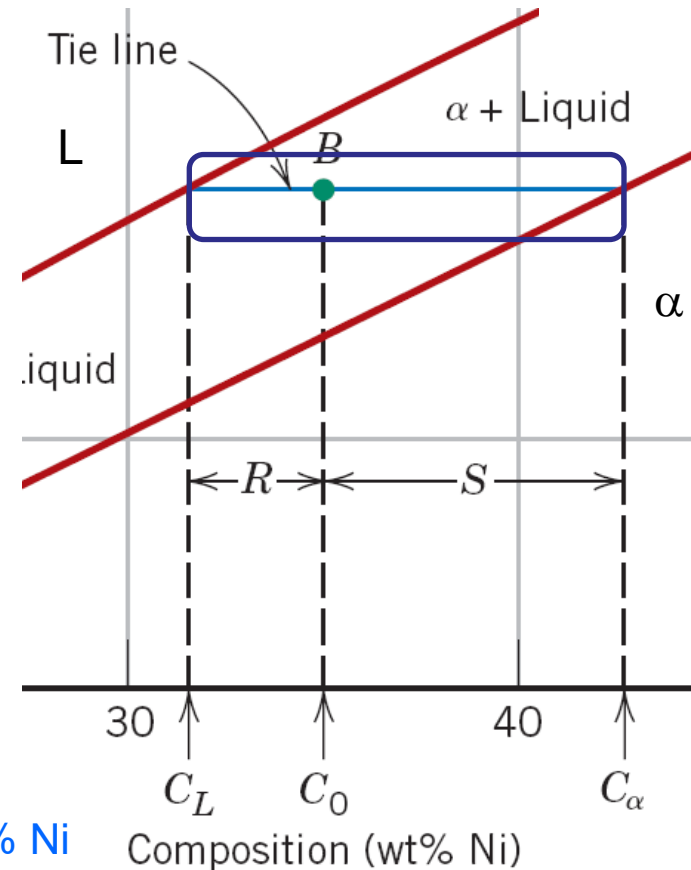
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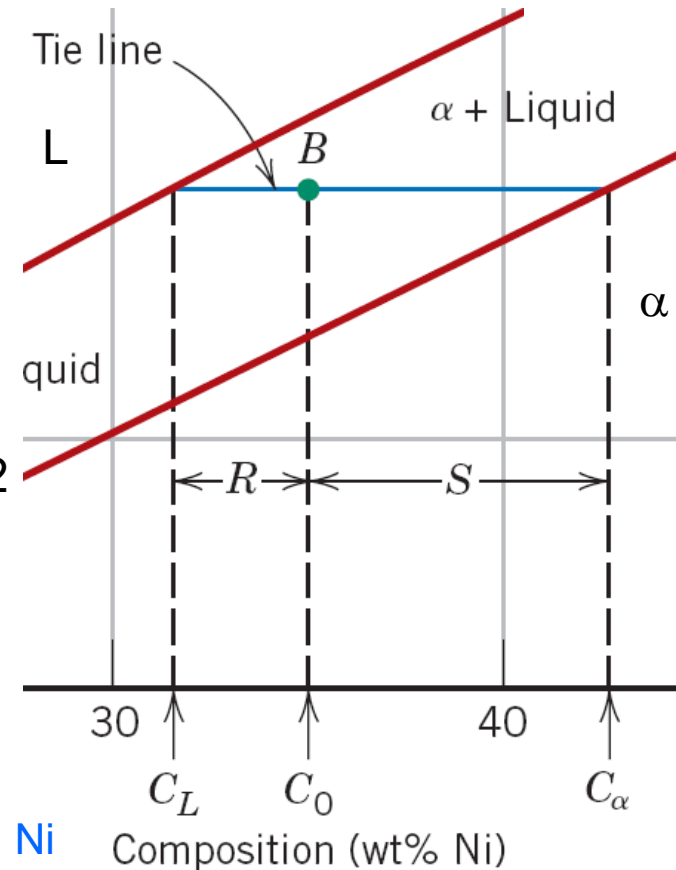
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$$W_\alpha = \frac{R}{R+S} = \frac{C_0 - C_L}{C_\alpha - C_L} = \frac{35 - 32}{43 - 32} = 0.28$$

$$W_L + W_\alpha = 1$$

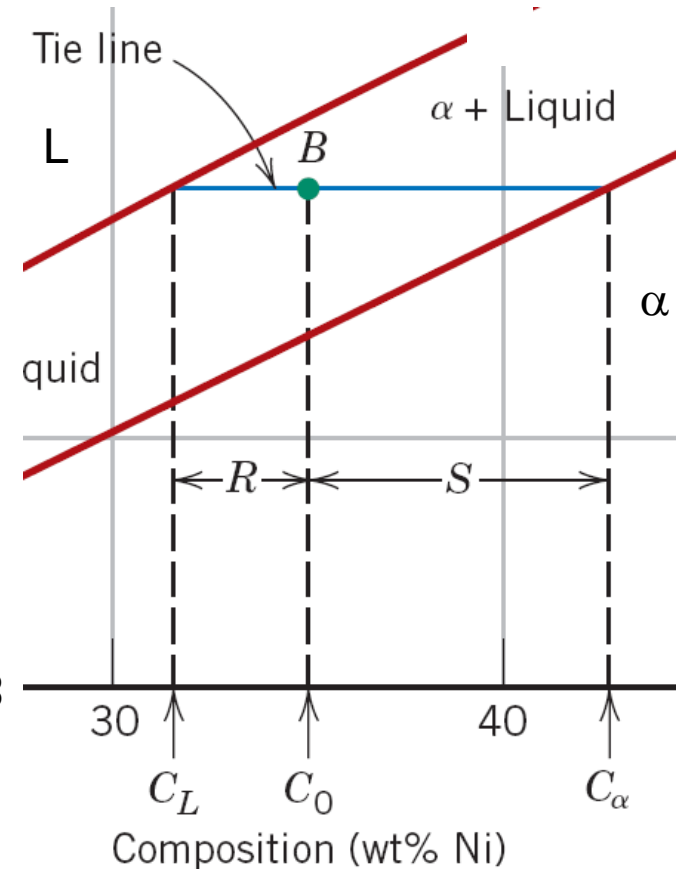
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NON-EQUILIBRIUM SOLIDIFICATION

Non-equilibrium solidification

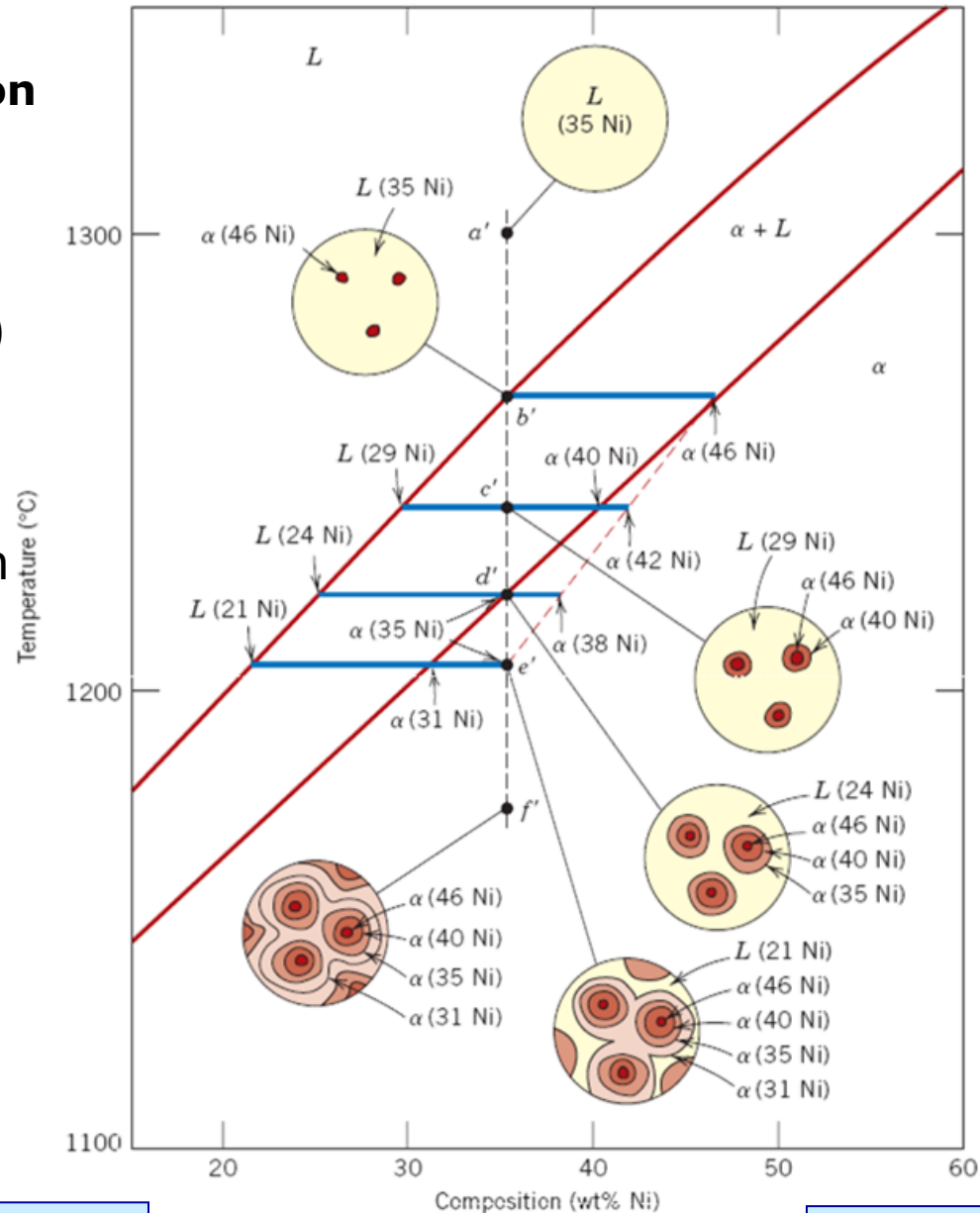
When we go down through the phase diagram too quickly (high cooling rates)



Microstructures of heterogeneous composition are produced

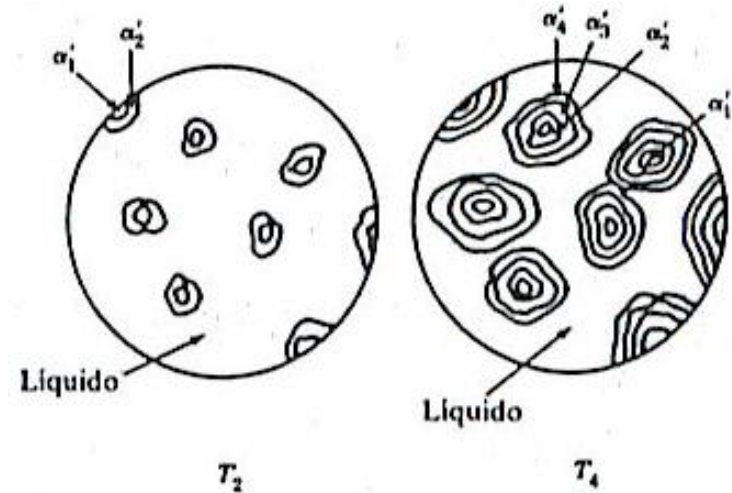
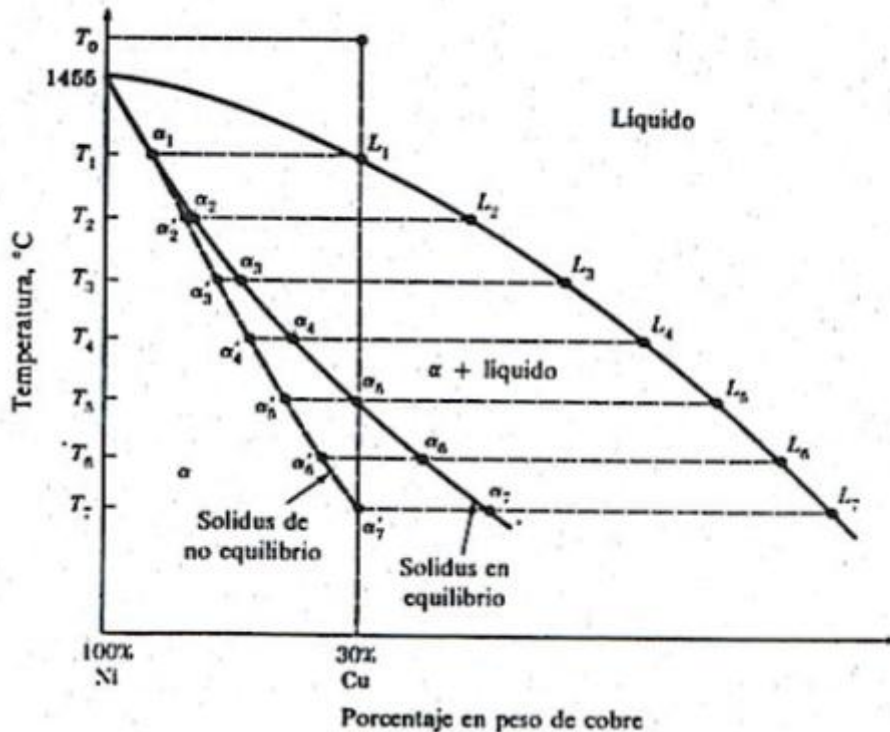


MICROsegregation (or cored structure)

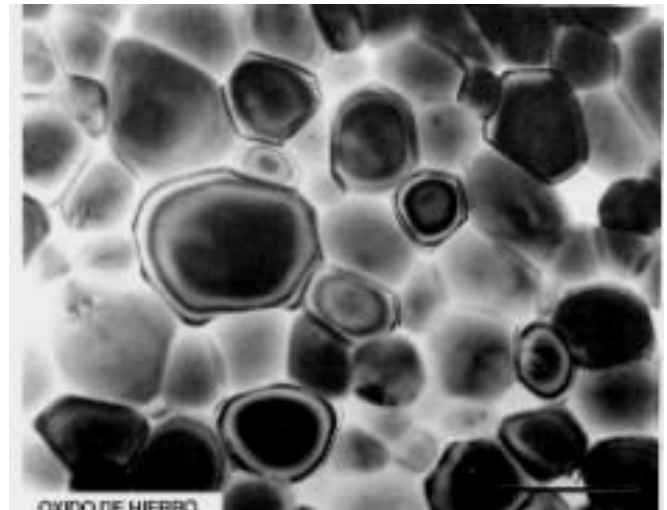


NON-EQUILIBRIUM SOLIDIFICATION

Non-equilibrium solidification



Microsegregation (or coring) can be eliminated with homogenization heat treatment at T below solidus (Example: $T_{TT} < T_7$). During heat treatment solid state diffusion occurs

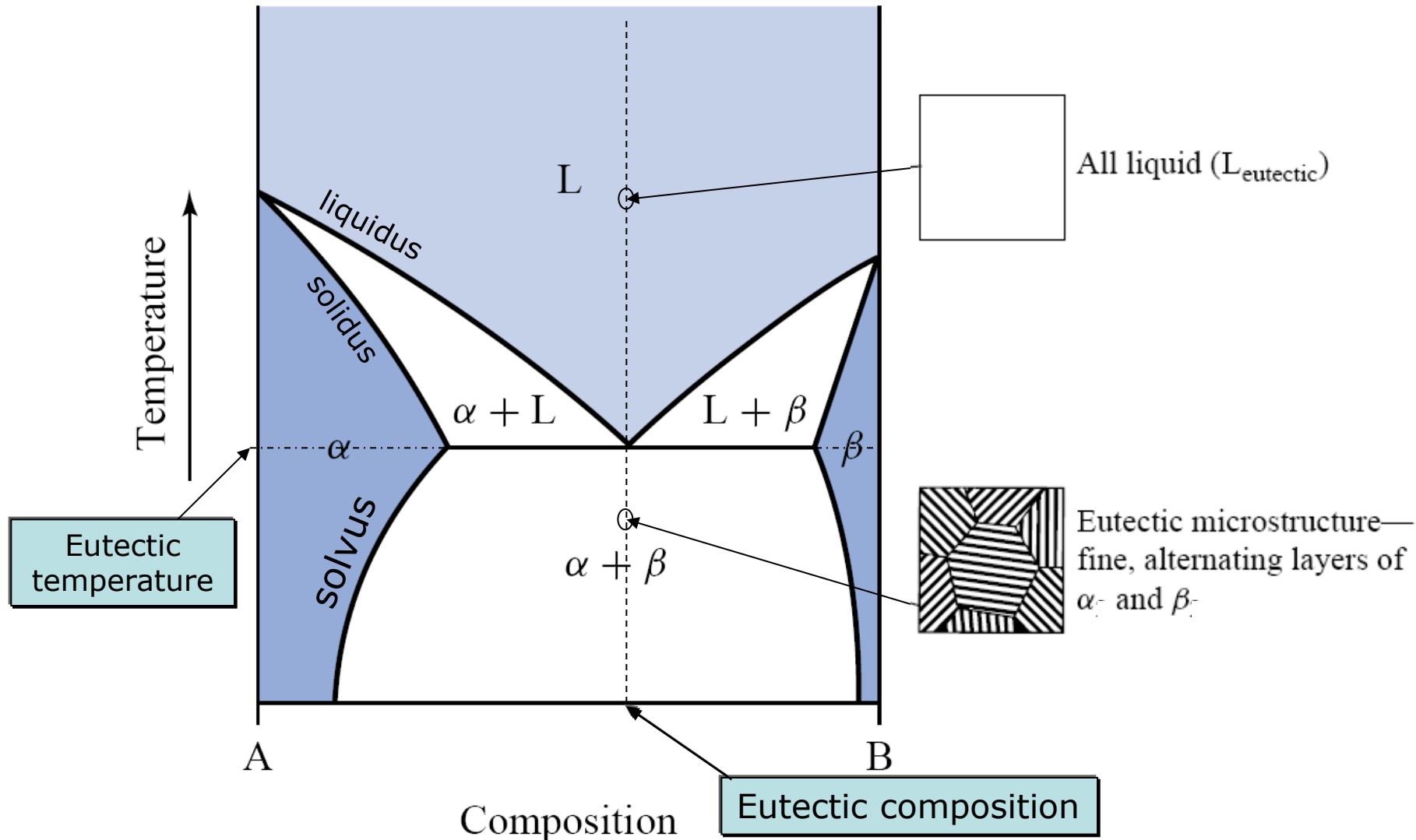


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BINARY EUTECTIC SYSTEMS

SYSTEMS OF 2 COMPONENTS WITH PARTIAL SOLID SOLUBILITY AND AN INVARIANT EUTECTIC REACTION

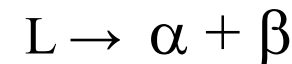


BINARY EUTECTIC SYSTEMS

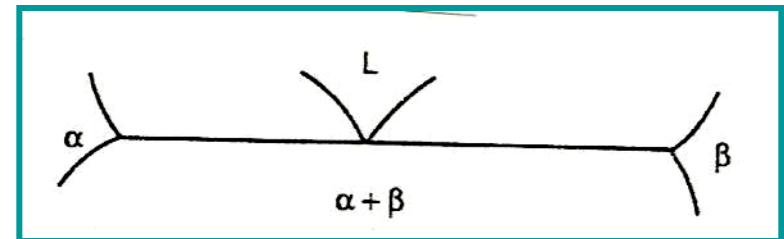
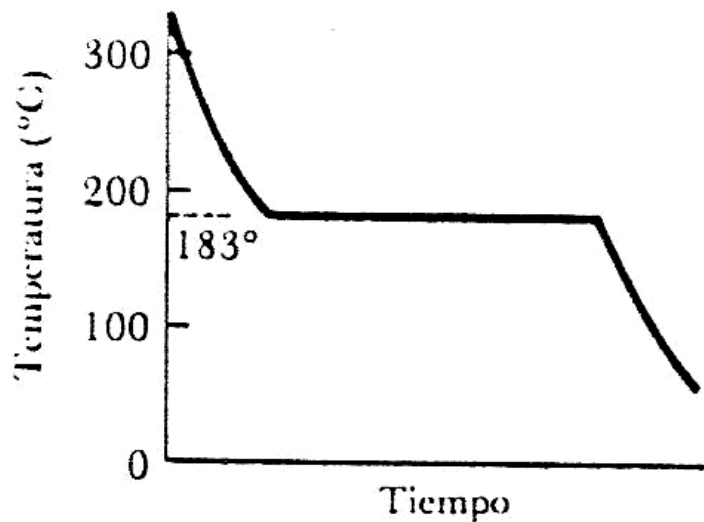
INVARIANT REACTIONS: They are produced at a fixed composition and temperature for a given system, and the result also remains fixed

In a system of two partially soluble components a solidification reaction is produced at a fixed composition and temperature with a profile of temperature gradient similar to pure solids.

Eutectic reaction

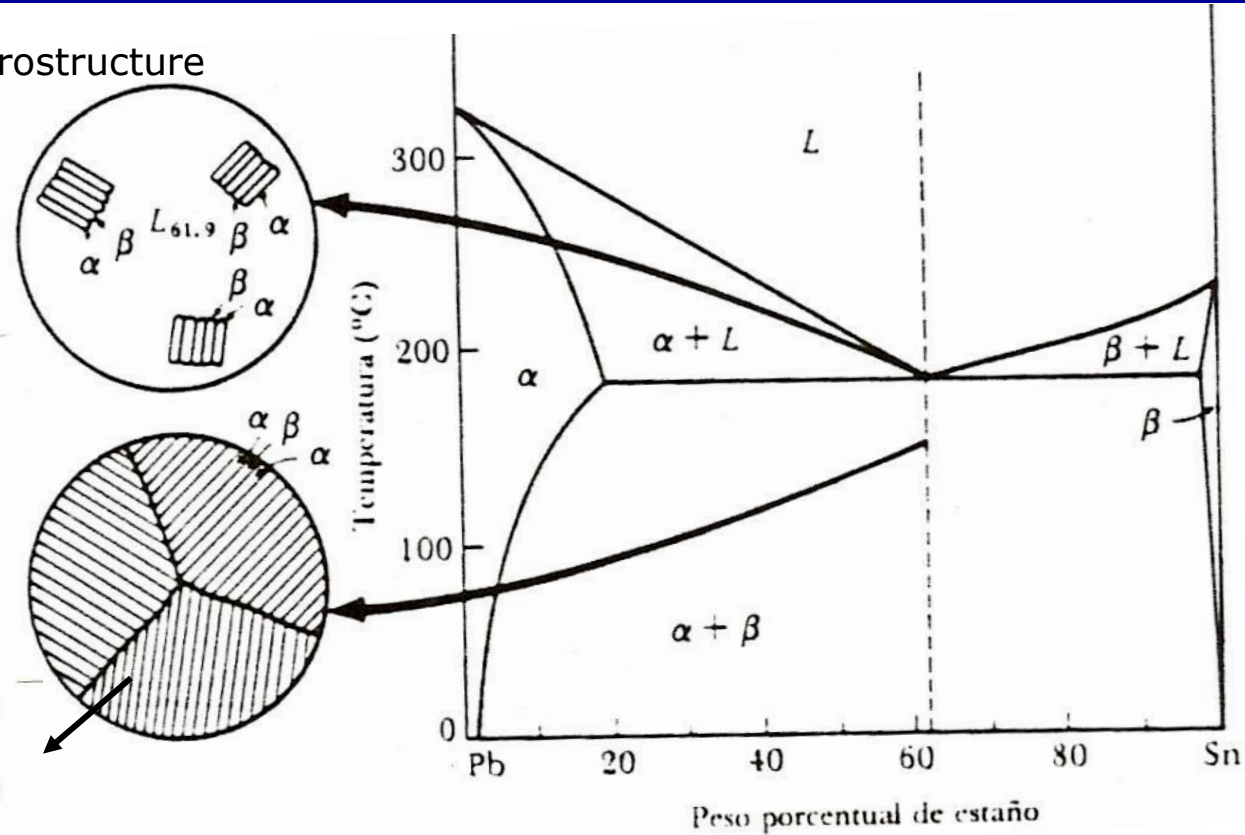


Cooling curve at eutectic Temperature

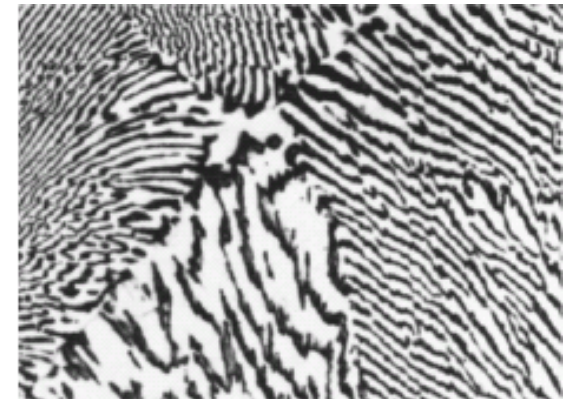
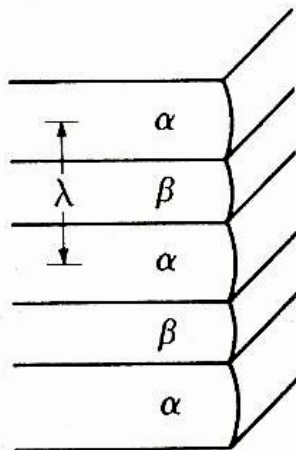


BINARY EUTECTIC SYSTEMS

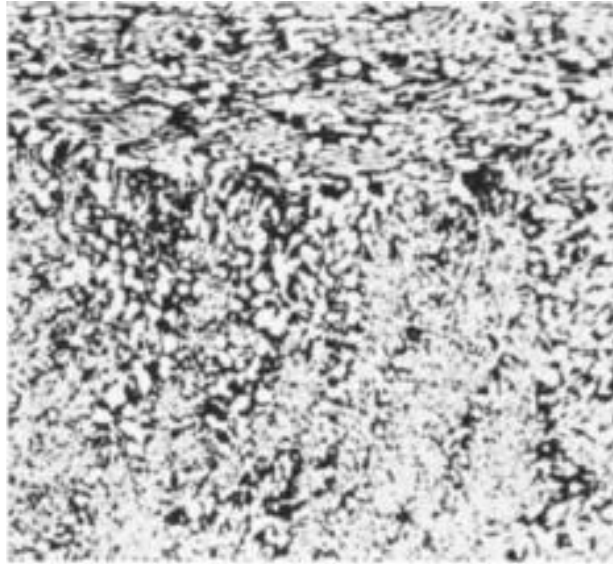
Eutectic reaction and microstructure



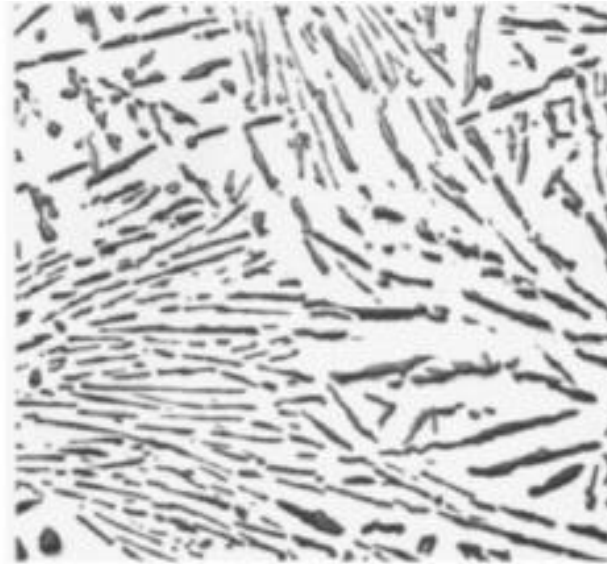
Eutectic product



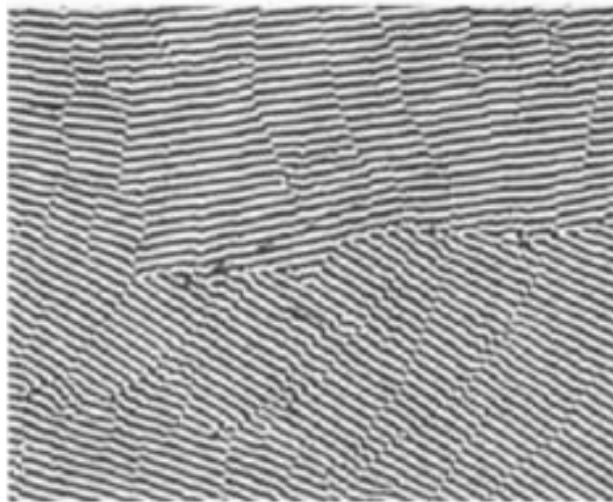
BINARY EUTECTIC SYSTEMS



(a)



(b)



(c)

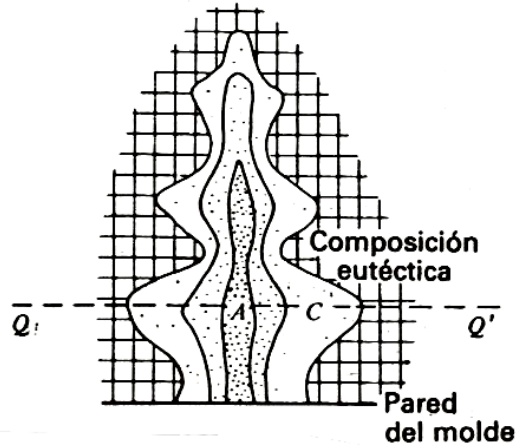


(d)

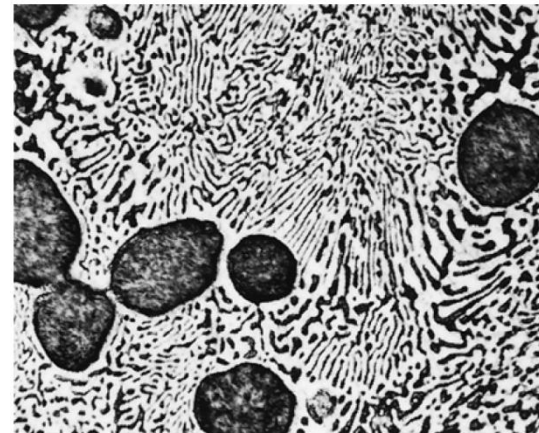
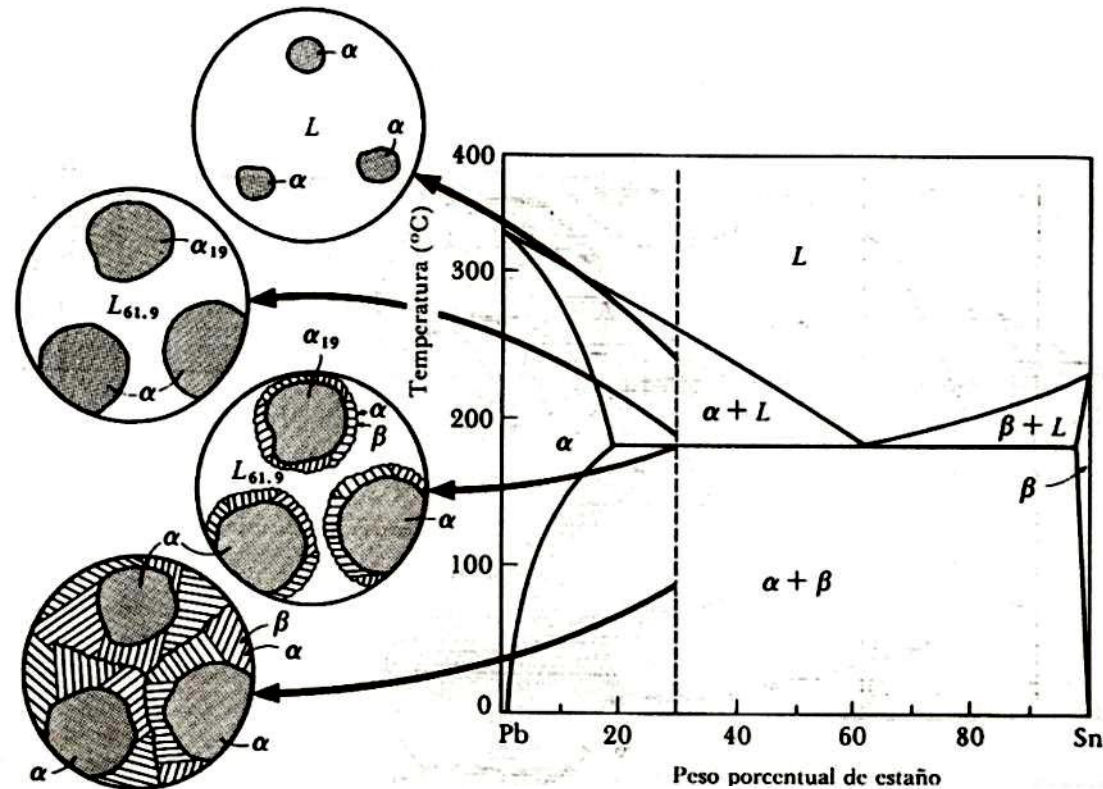
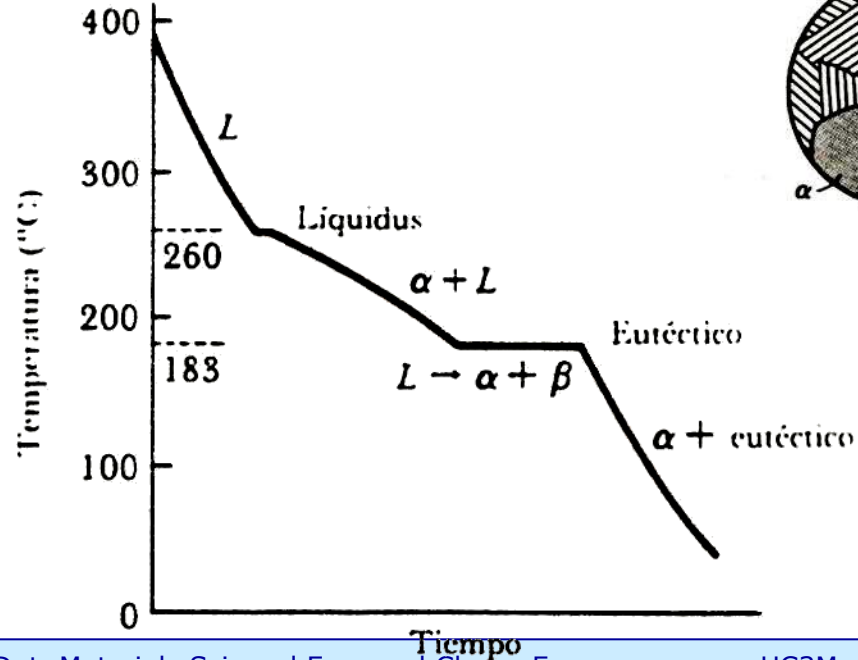
- a) Sn-50%In
- b) Al-13%Si
- c) Al-33%Cu
- d) Mg-37%Sn

BINARY EUTECTIC SYSTEMS

Hypo-Eutectic composition



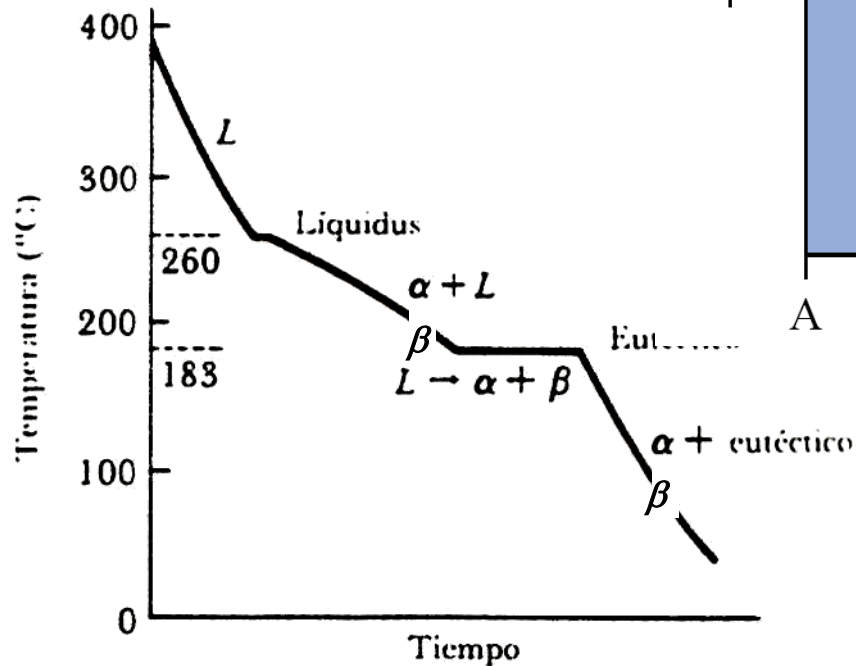
Hypo-Eutectic cooling curve



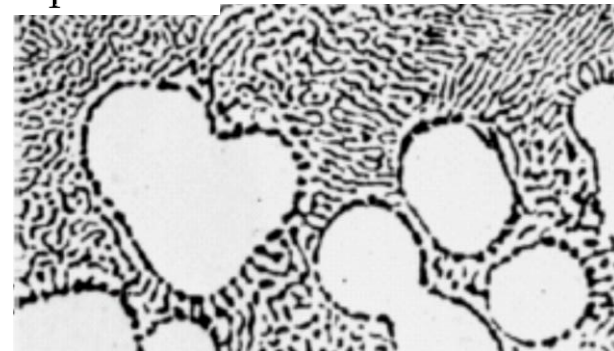
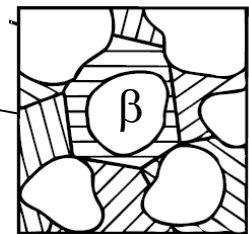
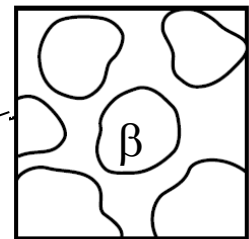
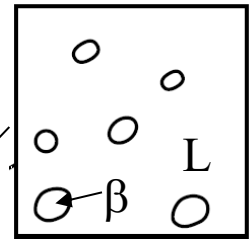
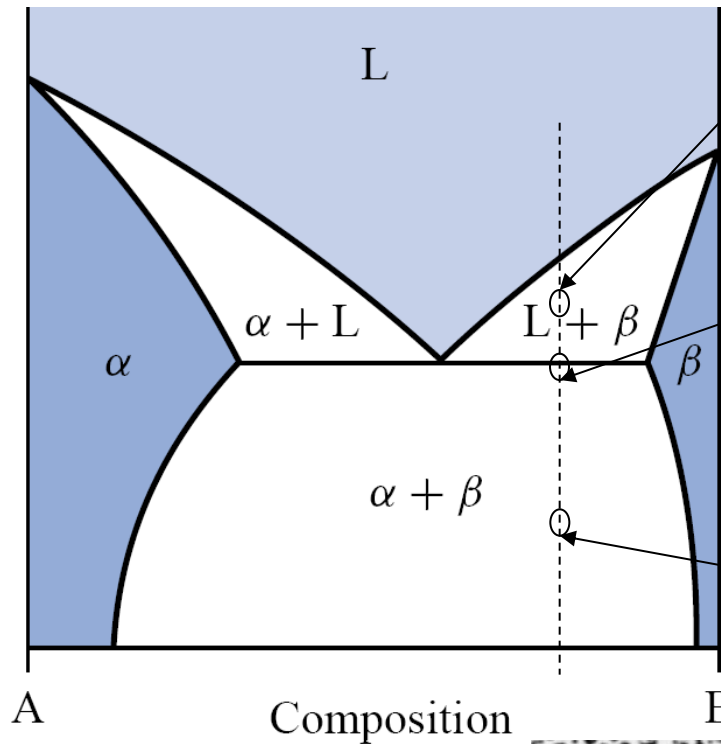
BINARY EUTECTIC SYSTEMS

Hyper-Eutectic compositions

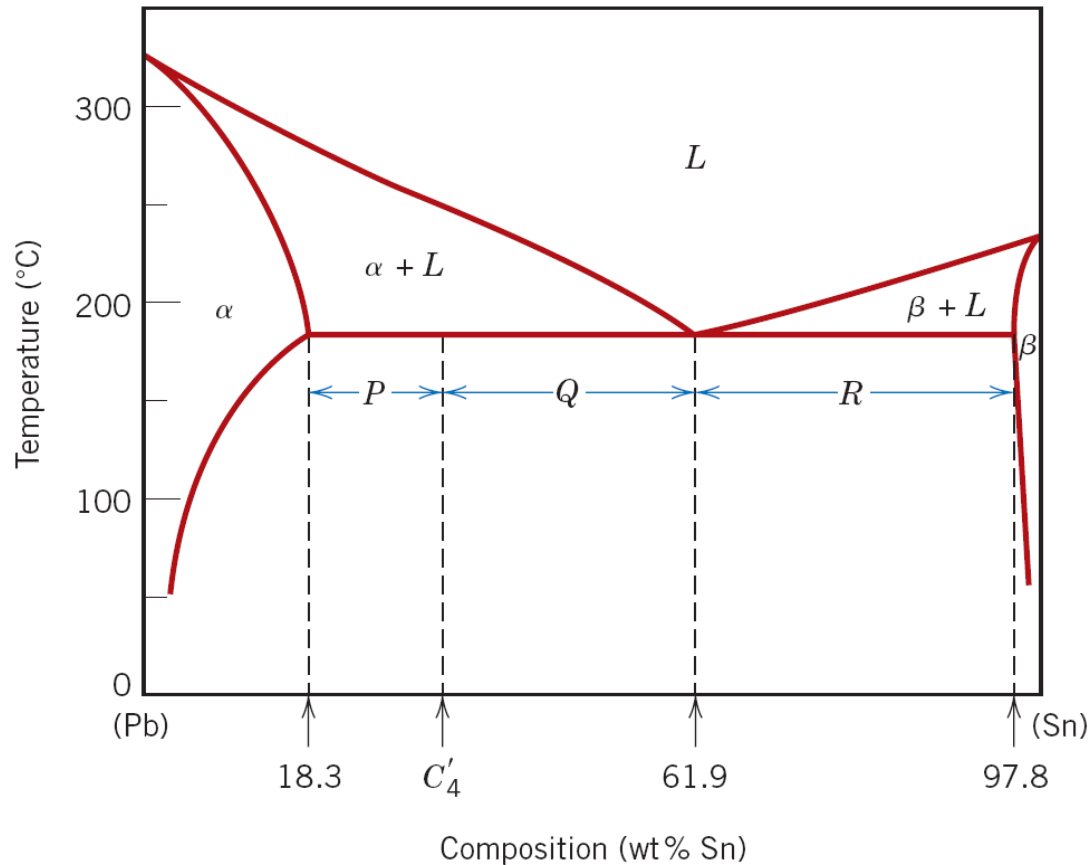
Hyper-Eutectic cooling curve



Temperature ↑



LEVER RULE IN BINARY EUTECTIC PHASE DIAGRAMS



$$W_{\alpha'} = \frac{Q}{P + Q}$$

$$W_{\alpha} = \frac{Q + R}{P + Q + R}$$

$$W_e = W_L = \frac{P}{P + Q}$$

$$W_{\beta} = \frac{P}{P + Q + R}$$

W_e = fraction eutectic microconstituent

W_L = fraction of liquid phase

$W_{\alpha'}$ = fraction of α' phase (primary α)

W_{α} = *total* fraction of α (primary α + α in eutectic)

W_{β} = fraction of β phase (total)